Cooper Wooley

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Education

Missouri University of Science and Technology

B.S. Computer Science, Mathematics Minor Dec 2025

GPA 3.84

Relevant Coursework: Machine Learning, Deep Learning, LLMs, Algorithms, Data Structures, Databases, Operating Systems, Networks, Security, Microcontrollers

Technical Skills

- **Languages:** Python, C/C++, JavaScript, SQL, HTML/CSS
- **Tools/Frameworks:** Linux, Docker, Git, VS Code, Arduino IDE

Experience

Formula Electric Design Team

Software Member Aug 2023 - Present

- Developed embedded software for various microcontrollers, including Teensy and ESP32 boards utilizing Arduino IDE and C++
- Enhanced data acquisition system with user-friendly visualizations to support mechanical and electrical sub-teams

City of Sikeston - Public Works, Parks and Recreation Sikeston, MO

Seasonal Skilled Worker May 2023 - Aug 2025 (Summers)

- Led field maintenance for baseball and soccer setups, ensuring optimal event preparation
- Coordinated daily field marking and equipment setup
- Participated in tornado and flood recovery efforts, safely clearing debris and restoring public spaces

Sikeston Public School District Sikeston, MO

IT Internship May 2022 – Aug 2022

- Established internal network infrastructure by configuring distribution frames
- Deployed switches and wireless access points; terminated fiber optic and Cat6 cabling
- Performed hardware troubleshooting and system reimaging

Projects

ParkVision (Capstone Project) – Python, OpenCV, Flask, PostgreSQL, Docker

- Engineered the backend for a smart parking lot dashboard using Flask and PostgreSQL, containerized via Docker
- Built and documented computer vision pipelines using OpenCV for automatic parking spot detection and background-based car occupancy analysis
- Developed REST API endpoints for initializing lots and retrieving live lot status using video input
- Implemented edge detection, reference frame analysis, and occupancy mapping for real-time data processing

CS2 Workshop Map Queue Bot – Python, Docker, Discord.py, Python-Valve

- Developed a Discord bot to manage 10-player CS2 custom games with automated queue and match systems
- Implemented ready checks, captain selection, and map veto logic for balanced team formation
- Automated Docker-based server spin-up and teardown with dynamic IP delivery to players